

Nikki Phach

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Work Experience

Konami Gaming Inc., Software Engineer I

April 2025 - present

- Enabled promotional free spin capability for 60+ online casinos by developing a universal feature that integrates with a third-party content provider's API and UI wrapper.
- Ensured compliance with Brazil gaming regulations by introducing a user confirmation flow for automatic spins within gameplay.
- Improved flexibility of RTP (return-to-player) configuration by redesigning the loading flow to externalize metadata and adding ability to override the RTP display on game launch.
- Released new game title, All Aboard Panda Pennies, by cloning an existing game and extending it with responsive orientation support, full-screen background handling, and character group animations for compatible with its unique large-reel feature.
- Prepared entry into the German market by developing methodology for treating progressive pools as static jackpots, using scripts to manipulate progressive databases and implementing a customizable RTP display.
- Delivered 10+ staging site launches and 70+ game deployments by managing site-specific configurations across multiple jurisdictions.
- Shipped 15+ game re-releases, resolving 85+ issues to ensure production stability.

Projects

Kotoba Tag!

kotoba-tag.com

- Implemented an answer validation pipeline with 95% accuracy by fine-tuning a SentenceTransformer model on 4M+ JMDict/WordNet gloss pairs for cross-lingual definition similarity.
- Curated a 5,800+ word Japanese vocabulary bank with JLPT filtering, part-of-speech validation and syllable matching logic.
- Deployed a production application with XState, React Router, and a responsive shadcn/ui layout.
- Built a FastAPI backend on Render proxying HuggingFace inference and Jisho.org lookups, with cold-start warmup and retry handling.
- Enhanced gameplay with scoring multipliers, configurable game settings, and dictionary detail lookup.
- Maintained reliability with a Vitest suite covering syllable matching, definition validation and game flow.

Personal Website

nphach.github.io

- Designed a memorable portfolio experience by crafting a retro gadget UI in React/TypeScript.
- Enabled seamless animations with responsive geometry, SVG masks, and Motion transitions.

Image Analysis: Skin Detection

Spring 2024

- Improved a skin detection pipeline from 73% to 98% accuracy by applying color-space analysis and dynamic thresholding, refactoring the core algorithm, and adding async image processing to reduce runtime.
- Built a ground-truth skin-mask test dataset with Photoshop and implemented a Metrics class to benchmark baseline performance and evaluate detection accuracy against labeled annotations.

Skills

TypeScript, JavaScript, Python, C, SQL, React, React Router, HTML/CSS, XState, TailwindCSS, shadcn/ui, Responsive Web Design, FastAPI, Flask, REST APIs, Phaser, Vitest, Unit Testing, Zod, npm, Docker, Git/GitHub, Linux/Unix, PyTorch, NumPy, Pandas, scikit-learn, HuggingFace, Machine Learning, Natural Language Processing, Computer Vision, Data Structures/Algorithms, Database Design, Data Processing

Education

University of Massachusetts Boston, B.Sc in Computer Science

May 2024

DeepLearning.AI, Stanford University, Machine Learning: Regression and Classification

January 2024